Airbnb Analytics Dashboard Project Report

Project Overview

The Airbnb Analytics Dashboard is an interactive Business Intelligence solution developed using Power BI to analyze Airbnb listings, host performance, property pricing, and customer review trends across multiple cities.

The dashboard transforms raw Airbnb listing and review data into meaningful business insights through advanced data cleaning, DAX calculations, Python visualizations, and interactive reporting.

The solution helps stakeholders answer key business questions related to:

- Pricing trends
- Host performance
- Customer satisfaction
- Property popularity
- Review growth patterns
- Revenue opportunities

Business Objective

The objective of this project is to:

- Analyze Airbnb listings across different cities.
- Evaluate host performance and Super Host behavior.
- Identify factors affecting customer satisfaction.
- Understand pricing trends across room types and property types.
- Analyze review growth and customer engagement.
- Provide actionable business insights for decision-making.

Tools & Technologies Used

Tool	Purpose
Power BI	Dashboard Development
Python	Data Cleaning & Python Visuals
Pandas	Data Manipulation
NumPy	Numerical Operations
DAX	KPI and Business Measures

Power Query	Data Transformation
Matplotlib	Python Visualizations

Dataset Information

Listings Dataset

Records

248,000+

Columns

33

Reviews Dataset

Records

5.3 Million+

Data Cleaning Process

The data cleaning process was performed in Jupyter Notebook using Python.

Steps Performed

Missing Value Treatment

Handled null values in:

- Bedrooms
- Review Scores
- Host Response Rate
- Host Acceptance Rate

using median imputation.

Duplicate Check

Checked and removed duplicate records.

Date Conversion

Converted:

- host_since
- review_date

to DateTime format.

DAX Measures

```
Accuracy Score =AVERAGE( Listings_Cleaned[review_scores_accuracy])
Average Price =AVERAGE( Listings_Cleaned[price])
Average Rating =AVERAGE(Listings_Cleaned[review_scores_rating])
Avg Acceptance Rate =AVERAGE(Listings_Cleaned[host_acceptance_rate])
Avg Accommodates =AVERAGE(Listings_Cleaned[accommodates])
Avg Bedrooms =AVERAGE(Listings_Cleaned[bedrooms])
Avg Listings per Host =DIVIDE( [Total Listings], [Total Hosts])
```

Avg Response Rate =AVERAGE(Listings_Cleaned[host_response_rate])
Check-in Score =AVERAGE(Listings_Cleaned[review_scores_checkin])
Maximum Price =MAX(Listings_Cleaned[price])
Property Listings =COUNTROWS(Listings_Cleaned)
Total Hosts =DISTINCTCOUNT(Listings_Cleaned[host_id)
Total Listings =DISTINCTCOUNT(Listings_Cleaned[listing_id])
Total Reviews =COUNT(Reviews_Cleaned[review_id])

Dashboard Pages

Page 1 – Executive Summary

Purpose

Provides a high-level overview of Airbnb business performance.

Visuals

- KPI Cards
- Listings by City
- Room Type Distribution
- Average Price by City
- Top Property Types
- Average Rating by Room Type

Key Insights

- Overall listings and reviews.
- Top-performing cities.
- Most popular room types.

Page 2 – Host Performance Analysis

Purpose

Analyze host behavior and performance.

Visuals

- Super Host Distribution
- Response Rate by City
- Acceptance Rate by City
- Top Hosts by Listings
- Host Verification Analysis

Key Insights

- Super Host contribution.
- Host responsiveness.
- Host acceptance trends.

Page 3 – Property & Pricing Analysis

Purpose

Analyze pricing and property characteristics.

Visuals

- Average Price by City
- Room Type Distribution
- Instant Bookable Analysis
- Property Type Analysis
- Bedrooms vs Price
- Accommodates vs Price

Key Insights

- Cities with highest prices.
- Most popular property types.
- Impact of bedrooms on pricing.

Page 4 – Review Analytics

Purpose

Evaluate customer satisfaction and review trends.

Visuals

- Average Rating by City
- Average Rating by Room Type
- Review Trend Analysis
- Rating Distribution
- Review Score Metrics

Key Insights

- Best-rated cities.
- Customer satisfaction levels.
- Review growth over time.

Page 5 – Python Analytics & Business Insights

Purpose

Perform advanced analytics using Python.

Python Visuals

Price Distribution Histogram

Used to identify:

- Pricing concentration
- Outliers
- Distribution patterns

Code

```
import matplotlib.pyplot as plt
plt.figure(figsize=(8,5))
plt.hist(dataset['price'], bins=30)
plt.title('Price Distribution')
plt.xlabel('Price')
plt.ylabel('Frequency')
```

```
plt.tight_layout()
plt.show()
```

Correlation Matrix

Used to analyze relationships between:

- Price
- Bedrooms
- Accommodates
- Ratings
- Location Scores

Code

```
import matplotlib.pyplot as plt
import pandas as pd
corr = dataset.corr(numeric_only=True)
fig, ax = plt.subplots(figsize=(8,6))
im = ax.imshow(corr)
ax.set_xticks(range(len(corr.columns)))
ax.set_yticks(range(len(corr.columns)))
ax.set_xticklabels(corr.columns, rotation=90)
ax.set_yticklabels(corr.columns)
plt.colorbar(im)
plt.title("Correlation Matrix")
plt.tight_layout()
plt.show()
```

Property Type Box Plot

Used to compare:

- Price distribution across property types
- Price variability

Code

```
import matplotlib.pyplot as plt
top_types = dataset['property_type'].value_counts().head(10).index
filtered = dataset[dataset['property_type'].isin(top_types)]
plt.figure(figsize=(10,6))
filtered.boxplot(
    column='price',
    by='property_type',
    rot=90)
plt.title('Price Distribution by Property Type')
plt.suptitle('')
plt.xlabel('Property Type')
plt.ylabel('Price')
```

```
plt.tight_layout()
plt.show()
```

Conclusion

The Airbnb Analytics Dashboard successfully transforms large-scale Airbnb listing and review data into meaningful business intelligence. Through interactive dashboards, DAX measures, Python visualizations, and advanced analytics, the solution enables stakeholders to monitor performance, identify trends, and make data-driven decisions.

Screenshots